Flow Measurement and Instrumentation 94 (2023) 102475

Contents lists available at ScienceDirect

Flow Measurement and Instrumentation

journal homepage: www.elsevier.com/locate/flowmeasinst

Parametric evaluation of compact truncated Venturi flow meters

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ARTICLE INFO

Keywords:
Venturi flow meter
Pressure loss
Discharge coefficient
Design of experiment
Differential pressure
Computational fluid dynamics

ABSTRACT

Venturi flow meters are pressure differential devices that perform accurate measurements with minimal pressure losses. They are widely used and well-documented in high-Reynolds applications. However, there is a lack of information regarding their performance outside the standards. The premise of the present study is to characterize a single-phase subsonic compact truncated venturi flow meter experimentally and then extend the research using Computational Fluid Dynamics (CFD). The computational work results in a design of experiment with 216 unique geometries at three different flow conditions (648 cases). This allows for characterizing the pressure losses and discharge coefficient as a function of diameter ratio (0.3–0.8), recovery cone truncation (0 %–50 %), and divergent angle (5°–20°). The thorough analysis of the results yields an intricate flow interaction and dependency on the studied parameters. The study covers throat Reynolds numbers from $7 \bullet 10^3$ to $7 \bullet 10^4$. In this regime, uncertainty in mass flow measurement increases notably (up to 20 %) with a decrease in mass flow. Pressure losses are sensitive to the three studied parameters, while the discharge coefficient can be considered to be independent of divergent angle and truncation. Truncating the Venturi at 50 % can increase the pressure loss coefficient by over 15 %. The optimal divergent angle and truncation to minimize losses for a prescribed length are provided for the instances where the installation space is constrained. This manuscript includes design guidelines relevant to all experimentalists.

1. Introduction

Venturi tubes are flow measurement devices that consist of an inlet cylinder, a conical convergent section, a cylindrical throat, and a conical divergent outlet section. This outlet (or recovery) cone is the main feature that notably reduces head losses compared to other pressure differential devices such as orifice plates or flow nozzles [1]. Lower pressure losses imply a reduction in the pumping power required for the line where the device is installed, thus lowering operating costs. The main guidelines for designing single-phase subsonic Venturi flow meters running full are given by the International Organization for Standardization (ISO) [2,3] and the American Society of Mechanical Engineers (ASME) [4]. These standards and numerous fluid mechanics books and handbooks [1,5–7] describe the Venturi flow meter design, installation, calibration, maintenance, and operation in detail.

Warren et al. [8] experimentally studied the head loss in Venturi meters, including the effects of diameter ratio, divergent angle, and truncation. They reported optimum recovery cone angles and concluded that Venturis with larger diameter ratios are less sensitive to a change in divergent angle. Additionally, they found that for a given divergent section length, truncating the Venturi (i.e., cutting off a portion of the recovery cone) may result in lower head loss compared to the full-length recovery cone with a larger angle. Sharp et al. [9] studied numerically (validating with experimental data) the optimal recovery cone angle of the classical Venturi to minimize head loss. This optimal angle varied widely (5.1°–12.2°) since it was found to be a strong function of the diameter ratio.

The behavior of the discharge coefficient, which relates the actual discharge to the theoretical discharge, depends on the geometry and Reynolds regime. At pipe Reynolds higher than $2 \bullet 10^5$, discharge coefficients are predictable for water applications. This is not the case for gases at high pressure and high Reynolds numbers [10]. Work by Reader-Harris et al. [11,12] explored the performance differences between water and gas. They also studied the effect of the convergent section shape on the device calibration. At low Reynolds, either at low speed or high viscosity, calibration can be challenging and lead to higher uncertainty [13–15]. Hollingshead et al. [16] numerically analyzed discharge coefficients of Venturi, orifice plate, V-cone, and wedge flow meters at low Reynolds numbers. In the investigated range, the discharge coefficient for Venturis presents a sharp decrease as the

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https://doi.org/10.1016/j.flowmeasinst.2023.102475
Received 22 July 2023; Received in revised form 28 September 2023; Accepted 11 October 2023
Available online 18 October 2023

Reynolds number decreases. They developed correction equations as a function of pipe Reynolds to predict the discharge coefficient performance.

Prior research lacks information regarding pressure losses and their sensitivity to geometrical parameters, especially with a scarcity of quantitative data regarding truncation. Venturi truncation is critical in many applications for all experimentalists due to spatial constraints in the existing hardware, particularly relevant in propulsion and the medical sector. The present manuscript attempts to fill this void in the literature. Furthermore, most of the documentation is focused on high-Reynolds. For instance, ISO 5167-4 [3] covers mostly pipe sizes between 50 mm and 1200 mm with Reynolds numbers higher than $2 \bullet 10^5$. Even though the designs in the present manuscripts follow some of the ISO guidelines, the diameter of the devices and the Reynolds are outside of the range covered by the standard. Venturi flow meters for low mass flows have aerothermal testing applications such as flow control (e.g., airflow injection to control stall or shockwaves [17]), heat transfer (e.g., tip coolant injection [18]), and spray injection (e.g., liquid jets in crossflow [19] and particle image velocimetry seeding [20]). Low Reynolds flows are also relevant in the biomedical field, including gases, e.g., assisted ventilation [21], and liquids, e.g., extracorporeal membrane oxygenation [22]. In the present work, the computational fluid dynamics (CFD) results are first validated with experimental data. Then, a design of experiment (DOE) including 216 geometries at three different flow conditions (648 cases) is calculated. Finally, the data is analyzed to report the sensitivity to diameter ratio, truncation, and divergent angle. Design recommendations are provided with a detailed analysis of the aerodynamic performance.

2. Methodology

2.1. Benchmark design and characterization of the test case

Fig. 1 shows the studied flow meter's schematic layout and measurement planes. The governing equation for the mass flow through a Venturi flow meter can be derived from continuity and momentum as [3]:

$$\dot{m} = \frac{C}{\sqrt{1-\beta^4}} \varepsilon \frac{\pi}{4} d^2 \sqrt{2 \Delta P \rho_{\text{in}}} \tag{1}$$

, where C is the discharge coefficient; $\beta = d/D$ is the ratio between the throat diameter, d, and the upstream diameter, D; ε is the expansibility factor; $\Delta P = P_1 - P_2$ is the differential pressure between the upstream (*Plane 1*) and the throat pressure tappings (*Plane 2*); and ρ_{in} is the density of the fluid at the inlet of the device. The density can be computed from the equation of state [2]:

$$\rho_{\text{in}} = \frac{P_{\text{in}}}{R T_{\text{out}}} \tag{2}$$

, where P_{in} is the pressure at the inlet plane (*Plane in*), T_{out} is the temperature at the outlet (*Plane out*), and R is the specific gas constant of the fluid running through the Venturi. This is computed under the assumption that the temperature does not change across the device, i.e., $T_{out} = T_{in}$.

The expansibility factor accounts for the compressibility of the fluid [3] and is computed as

$$\varepsilon = \sqrt{\left(\frac{\gamma \tau^{2/\gamma}}{\gamma - 1}\right)\left(\frac{1-\beta^4}{1-\beta^4 \tau^{2/\gamma}}\right)\left(\frac{1-\tau^{(\gamma-1)/\gamma}}{1-\tau}\right)} \tag{3}$$

, where γ is the gas specific heat ratio, and τ is the ratio between the throat and upstream pressures (P_2/P_1). This formula should only be applied for values of $\tau \geq 0.75$. For incompressible fluids, $\varepsilon = 1$. Pressure losses in the manuscript are presented as:

$$P_{\text{loss}} = P_{\text{in}} - P_{\text{out}} - \left(P_{\text{in,pipe}} - P_{\text{out,pipe}}\right) \tag{4}$$

, where the pipe losses prior to the installation of the Venturi, $P_{\text{in,pipe}} - P_{\text{out,pipe}}$, are subtracted. This value is commonly found in the literature [3,4] as a pressure loss ratio:

$$\xi = \frac{P_{\text{loss}}}{\Delta P} \tag{5}$$

Losses are also reported as a pressure loss coefficient:

$$K = \frac{P_{\text{in}} - P_{\text{out}}}{\frac{1}{2}\rho_2 |v_2|^2} \tag{6}$$

, where the throat density, ρ_2, and the throat absolute velocity, $|v_2|$, are used to compute the dynamic pressure component.

Other parameters used to define the geometry are the convergent angle of the entrance cone, α_c, the throat length, L_{throat}, and the divergent angle of the recovery cone, α_d, as shown in Fig. 1. To shorten the length of the flow meter, the divergent section can be truncated so that the flow expands to the pipe diameter D with a step change in area. In this manuscript, we define a truncation parameter, $\theta = 1 - L_d/L_{d,0}$, where L_d is the divergent section length after truncation and $L_{d,0}$ is the full recovery cone length. The venturi divergent section length can then be computed using the equation:

$$L_d = \frac{D-d}{2\tan\left(\frac{\alpha_d}{2}\right)}(1-\theta) \tag{7}$$

To study the design effects for compact Venturis, a benchmark geometry was designed and manufactured at the Purdue Experimental Turbine Aerothermal Laboratory (PETAL). This test article provides accurate measurements for gas at low mass flows, generating a database for CFD validation. The flow meter design, shown in Fig. 1, follows some of the classical Venturi guidelines given by ISO 5167-4 [3]. However, the small diameter and low Reynolds number fall outside the bounds specified by the ISO. The device is 122 mm in length and 27 mm in outer diameter. The material chosen was brass since it provides easy

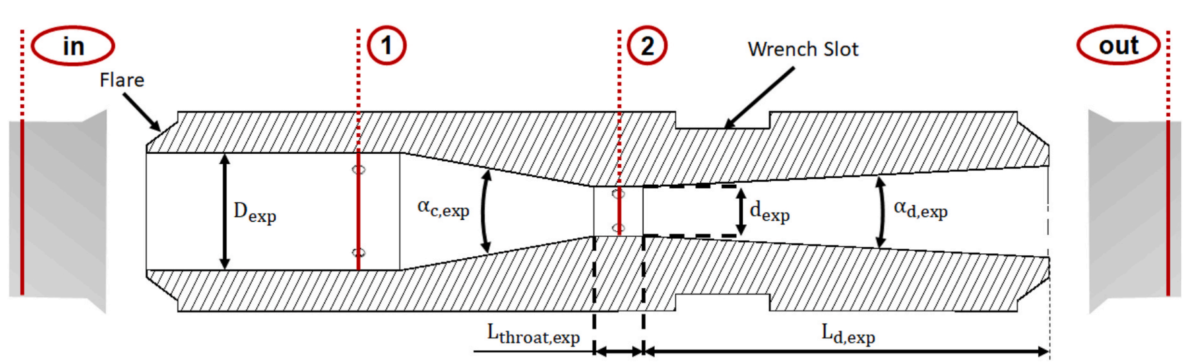

Fig. 1. Benchmark Venturi flow meter sketch of features and plane definitions.

machinability and superior corrosion resistance. Both ends are threaded and flared for a tight sealing to the piping while allowing easy assembly and operation.

There are two measurement planes on the flow meter, 5.6 mm upstream of the convergent section (*Plane 1*) and the other centered at the throat (*Plane 2*), as indicated in Fig. 1. Both planes have four equidistant cylindrical pressure tappings (1.6 mm in diameter), spaced 90° in the circumferential direction. The convergent section angle $\alpha_{c,exp}$ is 20° and the divergent section angle $\alpha_{d,exp}$ is 6°, both close to recommended values [3,4]. A shorter length was achieved by truncating the downstream section. ISO 5167-4 [3] states that a truncation of up to 35 % does not significantly modify pressure loss or discharge coefficient. To assess the effect of truncation, a higher value of 40 % ($\theta = 0.4$) was selected. Due to the high sensitivity of the mass flow calculations to changes in the throat section area, the diameter was measured using a Mitutoyo 2109S-10 dial indicator after manufacturing. The rest of the measurements provided were obtained using a Digital Electronic Caliper. The upstream diameter D_{exp} is 16.13 ± 0.02 mm, and the throat diameter d_{exp} is 6.754 ± 0.005 mm. The length of the throat $L_{throat,exp}$ is 7.88 ± 0.02 mm. The resulting diameter ratio β_{exp} is 0.42, which is in the range between 0.4 and 0.75 recommended for Venturi tubes with machined convergent sections on pipe sizes between 50 mm and 250 mm [3].

2.2. Numerical setup

The flow meter geometry is initially parametrized in Matlab [23], where the parameters shown in Fig. 2a can be tuned to generate a 3D model (Fig. 2b). An unstructured computational grid is generated for the domain using Hexpress [24]. Finally, the Reynolds Averaged Navier-Stokes (RANS) equations are solved with ANSYS-Fluent [25] on the computational domain.

Constant lengths are prescribed upstream and downstream of the convergent section inlet, 75 mm and 240 mm, respectively. The domain is discretized using a second-order upwind Finite Volume Method, and the turbulence closure is achieved with the $k-\omega$ SST model. The first cell center in the wall-normal direction is positioned such that $y^+ < 1$. The generated grids had 2.5 million cells on average. For the boundary conditions, mass flow and total temperature are prescribed at the inlet, and static pressure is prescribed at the outlet, as shown in Fig. 2b. At the inlet plane, the turbulent intensity is set to 5 %, and the ratio of turbulent viscosity to molecular viscosity is set to 10. The rest of the domain boundaries are set to adiabatic no-slip walls. The walls are modeled as a smooth surface, and the joints between the different sections of the Venturi tube are modeled as sharp corners. Four different planes are monitored for postprocessing: inlet, outlet, 5.6 mm upstream of the convergent section (*Plane 1*), and throat (*Plane 2*). Mass-flow averaged flow quantities are reported for each plane.

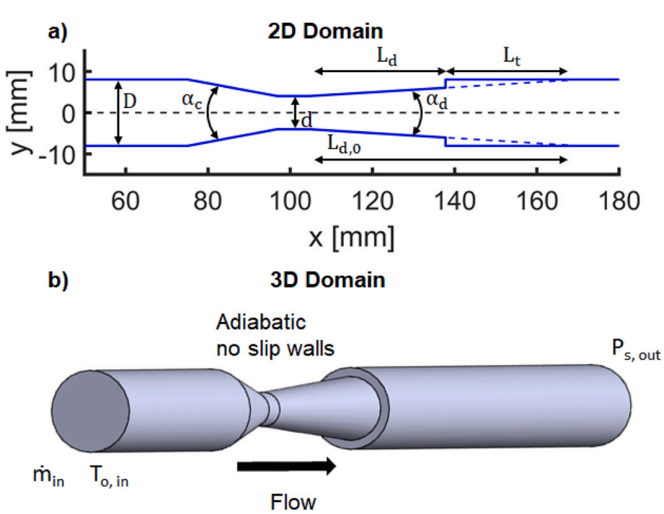

Fig. 2. a) Parametrization of the computational domain. b) Three-dimensional domain and boundary conditions used for the CFD.

A mesh sensitivity analysis was performed to confirm the grid independence of the results. The medium mass flow case (3.1 g/s) with a diameter ratio of 0.4, 30 % truncation, and 8° at the divergent section (*Model 56* in the Design of Experiment) was used. The convergent angle, α_c, was set to 21° with the inlet diameter D equal to 16.13 mm. A value of $L_{throat} = d$ was prescribed, where d was computed based on the diameter ratio, β, and the inlet diameter, D. Eight different meshes were generated in Hexpress with uniformly varying numbers of cells in the axial, radial, and azimuthal directions while keeping the first cell thickness constant and equal to 0.001 mm ($y^+ < 1$). The boundary layer cell stretching ratio was set to 1.2. Two quantities of interest were studied for each of these runs: pressure loss ratio (Equation (5)) and differential pressure between *Plane 1* and *Plane 2*. The mesh was refined from approximately 0.2 to 3.3 million cells (see Fig. 3). The difference between the selected grid (2.7 million cells) and the finest grid is 0.82 % and 0.08 % for the pressure loss ratio and the differential pressure, respectively. The grid selected provided a good balance between accuracy and computational cost, which was required due to the large number of cases.

2.3. Experimental setup

The experimental setup shown in Fig. 4 was designed to accurately characterize the benchmark Venturi flow meter under different mass flow conditions. Upstream of the facility, a pressure regulator controlled the flow of nitrogen into a CMFS025M Coriolis Flow Meter (0.25 % accuracy) from Emerson. The rest of the circuit was connected using 3/4 in. OD tubing with 0.065 in. wall thickness. The computation of pressure losses (Equation (4)) requires static pressure measurements upstream and downstream of the device, while the density correction (Equation (2)) requires a temperature measurement. The inlet pressure tap was located 191 mm upstream of the Venturi. The outlet pressure tap and the outlet thermocouple were located 203 mm and 686 mm downstream of the Venturi, respectively. These pressure taps and the flow meter measurement plane taps were connected to a 50 psi (±0.06 % full-scale accuracy) Scanivalve MPS4264 pressure scanner. The K-type thermocouple was read from a VTI EMX-1048A acquisition unit and was previously calibrated in a Fluke 9171 metrology well, which helped to increase the accuracy to 0.5 K. The pressure was increased from the upstream regulator in steps until stable nitrogen mass flows were achieved for 11 different conditions (from 0.6 g/s to 5.6 g/s). Throughout the test, the upstream pressure, P_{in}, ranges from 99.59 kPa to 102.59 kPa. The temperature, T_{out}, ranges from 299.3 K to 295.9 K.

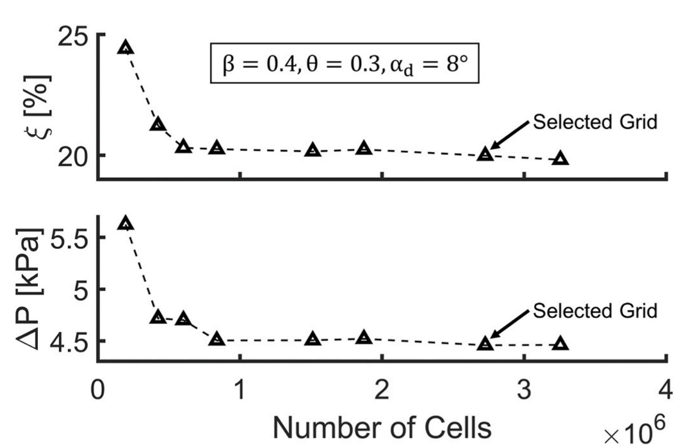

Fig. 3. Mesh sensitivity analysis for *Model 56*.

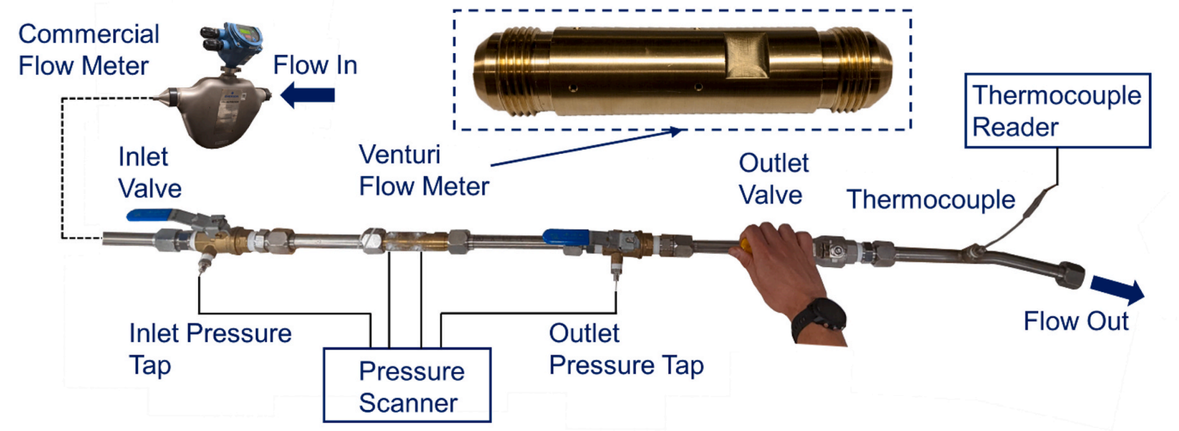

Fig. 4. Experimental setup.

2.4. Design of experiment (DOE)

The Design of Experiment (DOE) aims to explore how three geometric variables (viz., diameter ratio, truncation, and divergent cone angle) drive pressure losses and discharge coefficient in a Venturi flow meter. To study the interaction between these variables, a full factorial approach was implemented. The design variables were the diameter ratio, $\beta = d/D$, which was changed from 0.3 to 0.8 in increments of 0.1; the truncation ratio, $\theta = (1 - L_d / L_{d,0})100$, which was changed from 0 % to 50 % in increments of 10 %; and the divergent cone angle, α_d, which was changed from 5° to 20° in increments of 3°. This results in a DOE with $6^3 = 216$ unique geometries. The parameter values for each particular geometry (labeled as *Model 1* to *Model 216*) are reported in Appendix Table A 1. The convergent angle, α_c, was left constant and equal to 21° [3,4]. The inlet diameter D was also kept constant and equal to 16.13 mm. A value of $L_{\text{throat}} = d$ is prescribed for all the generated individuals.

The DOE was run for three different mass flow conditions tested in the experiment (section 2.3): low (1.2 g/s), medium (3.1 g/s), and high (5.6 g/s). In total, 648 unique CFD runs were produced. To automate the process, a wrapper for the geometry generation, meshing, and CFD solving was coded in Matlab [23]. The routine was executed in a high-performance computing (HPC) cluster where the different program interactions were handled via Bash scripts. A flow chart of this sequence is depicted in Fig. 5. Every time a CFD run was completed and post-processed, the results were saved in a database, and the next DOE variables were fed to the algorithm for a new computation.

To further assess the effects of the parameters, a quadratic response surface was fitted to the pressure loss data after excluding individuals with unsteadiness or sonic conditions. The approximation (surrogate model) of the function takes the form [26]:

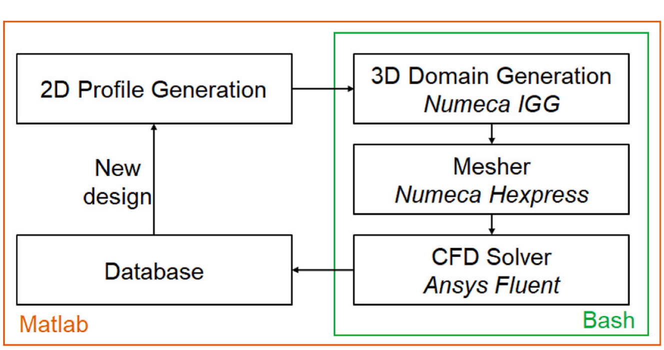

Fig. 5. Overview of the automated computational design of experiment.

$$\hat{f}(\boldsymbol{x}) = a_0 + \sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{n} \left[\sum_{j=i}^{n} a_{ij} x_i x_j \right] \tag{8}$$

, where the number of design variables is $n = 3$, the design variable vector is $\boldsymbol{x} = [x_1 \quad x_2 \quad x_3]^{\mathrm{T}} = [\beta \quad \theta \quad \alpha_d]^{\mathrm{T}}$, and a_0, a_i, a_{ij} are the quadratic approximation coefficients.

3. Results and discussion

3.1. Experimental validation

Experimental validation of the computational flow model was carried out by comparing the benchmark design performance to numerical results. The numerical boundary conditions were adapted to each of the 11 mass flow points tested experimentally.

The differential pressure was extracted at the same location from the CFD and the experiment model pressure taps for each recorded mass flow. This is shown in Fig. 6a. The data shows good matching between the two. Similarly, the pressure losses are compared in Fig. 6b. To isolate the pressure losses due to the flow meter, the experiment was run before and after the installation of the Venturi tube for the same 11 mass flow cases. The head losses caused by the pipe and fittings were subtracted from the total Venturi losses. As shown, the experiment and CFD data match with a maximum difference of 35.4 % at the lowest mass flow run and a minimum difference of 0.6 % for the medium mass flow run. The CFD underpredicts the pressure losses for the first 5 cases and estimates higher pressure losses for the higher mass flow cases.

The discharge coefficient, C, was computed from the experimental data and the CFD. The experimental discharge coefficient is obtained by solving Equation (1) for C, using the mass flow obtained from the reference Coriolis meter. The values are reported in Table 1, together with the corresponding Reynolds number and uncertainty. The pipe Reynolds number can be computed for each test point as $\mathrm{Re}_1 = \rho_1 |v_1| D / \mu$ and the throat-based Reynolds number is defined as $\mathrm{Re}_2 = \mathrm{Re}_1 / \beta$. The uncertainty in discharge coefficient is calculated using the sensitivity and measurement uncertainty approach presented by Huber et al. [27]. It is reported for a 95 % confidence interval (CI). The discharge coefficients obtained at the largest mass flow conditions are consistent with the 0.970 provided by the ISO [3] for Venturi tubes with a machined convergent section at $\mathrm{Re}_2 = 5 \bullet 10^4$.

A sensitivity analysis was carried out for each measured quantities to analyze the uncertainty further [27]. Sensitivity is defined as a percentage change in the discharge coefficient for a percentage change in input quantity. For a function $C = f(X)$, sensitivity of C to X is defined as

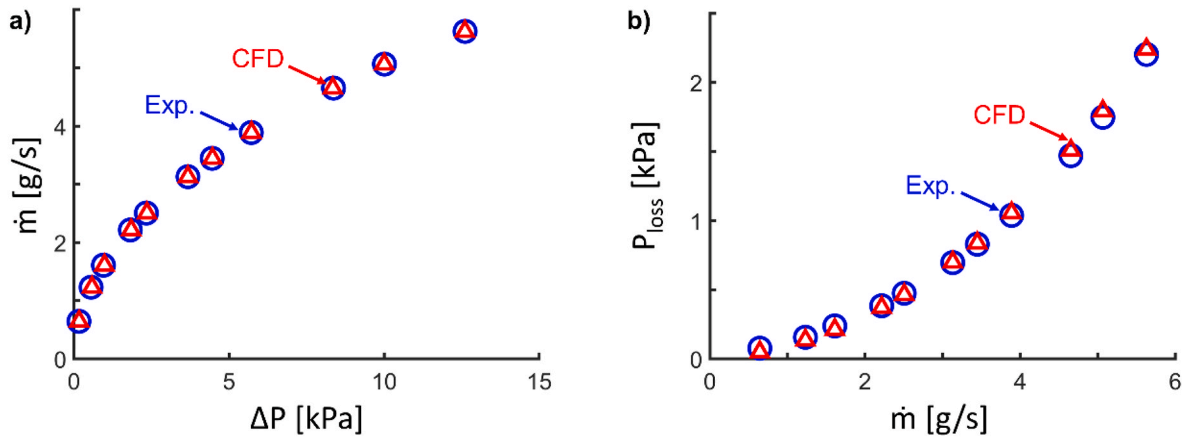

Fig. 6. Comparison between the experimental data and the numerical results.

Table 1
Discharge coefficients from experiment and CFD.

$\dot{m}$ [g/s]	Re_1	Re_2	C_{exp}	95 % CI	C_{CFD}
0.643	3035	7248	0.939	0.45	0.922
1.231	5644	13479	0.966	0.20	0.941
1.609	7402	17677	0.963	0.13	0.947
2.215	10199	24355	0.962	0.07	0.955
2.505	11537	27551	0.962	0.06	0.958
3.130	14399	34386	0.963	0.04	0.962
3.447	15814	37764	0.966	0.03	0.964
3.890	17820	42554	0.967	0.02	0.966
4.653	21303	50874	0.968	0.02	0.968
5.066	23144	55269	0.970	0.01	0.969
5.628	25679	61323	0.971	0.01	0.970

$$S(X) = \frac{\frac{\Delta C}{C}}{\frac{\Delta X}{X}} \tag{9}$$

Sensitivity is calculated for each input parameter and mass flow condition, as shown in Table 2. It can be seen that the discharge coefficient is directly dependent on the mass flow measurement, but the dependence remains constant, as well as for temperature. The dependence of throat diameter and inlet diameter changes weakly with mass flow. The discharge coefficient is most dependent on measured pressures at Plane 1 and Plane 2 and their changes with mass flow. The pressure measurements are used to compute the expansibility factor (ε), density (ρ_{in}), and differential pressure (ΔP), the latter having the most important influence on the sensitivity. At low mass flows, the dynamic pressure component that yields a differential pressure between P_1 and P_2 is greatly reduced, this implies that the measurement uncertainty is very high for low mass flows.

Table 2
Sensitivity (in %) of Discharge Coefficient to measured quantities.

Measured $\dot{m}$, g/s	Quantities					
	$S(d)$	$S(D)$	$S(P_1)$	$S(P_2)$	$S(T)$	$S(\dot{m})$
0.643	203.3	6.2	30294.1	30143.7	49.9	100.0
1.231	203.3	6.1	19440.4	19137.8	49.9	100.0
1.609	203.3	6.1	14538.7	14150.1	49.9	100.0
2.215	203.2	6.0	9498.0	9009.2	49.9	100.0
2.505	203.1	6.0	7890.2	7367.2	49.9	100.0
3.13	203.0	5.9	5499.5	4924.6	49.9	100.0
3.447	202.9	5.8	4680.5	4088.2	49.9	100.0
3.89	202.8	5.7	3778.8	3168.3	49.9	100.0
4.653	202.6	5.5	2699.2	2070.3	49.9	100.0
5.066	202.5	5.3	2299.1	1665.6	49.9	100.0
5.628	202.2	5.1	1866.0	1230.2	49.9	100.0

3.2. DOE results

The performance of a flow meter is assessed through pressure loss and discharge coefficient. The pressure losses are presented as a pressure loss coefficient, K, computed from Equation (6). The discharge coefficient, C, is computed from Equation (1). All the values for each individual geometry are reported in Table A 1 in the Appendix. The pressure loss ratio, ξ, computed from Equation (5), is also included in the table for completeness. Since throat Reynolds number and throat Mach number mainly depend on the diameter ratio, the average values (for different truncations and divergent angles) are reported in Table 3. The corresponding pipe Reynolds number (Re_1) is the same one that is listed in Table 1 for the experiment.

The pressure loss coefficients obtained for each of the 648 cases are shown in Fig. 7, with different diameter ratios, truncation, and divergent angles for different mass flows. There is a large scatter in the pressure losses for the smallest throat cases with high mass flow ($\beta = 0.3$ and $\dot{m} = 5.6$ g/s), as the flow reaches sonic conditions. This is seen in the contour of the Mach number in Fig. 8a. Since critical flow Venturi nozzles (operating under choked conditions) are beyond the scope of this manuscript, those individuals were not considered for further post-processing. Similarly, a large scattering is present for most of the cases with divergent angles larger than 11° ($\alpha_d = 14°$, $\alpha_d = 17°$ and $\alpha_d = 20°$). This is due to the unsteadiness generated by a large separation in the divergent section. Unsteady RANS computations would be required to average the pressure losses of these individuals more accurately. Fig. 8b shows the unsteady Mach contour for one such case. The angle at which this occurs ($\alpha_d = 14°$) is consistent for most data sets except for very large diameter ratio cases ($\beta = 0.8$ in Fig. 7). This finding agrees with the ASME Performance Test Codes [4] stating that unsteady pressure differentials are to be expected for cone angles exceeding about 15°. Additionally, the low mass flow ($\dot{m} = 1.2$ g/s) data set for low diameter ratios (i.e. $\beta = 0.3$ and $\beta = 0.4$ in Fig. 7) presents unsteadiness earlier on at $\alpha_d = 11°$.

The discharge coefficients from the DOE are shown in Fig. 9. The

Table 3
Averaged throat Mach number and throat Reynolds number for each diameter ratio and mass flow included in the DOE.

β	$\dot{m} = 1.231$ g/s		$\dot{m} = 3.130$ g/s		$\dot{m} = 5.628$ g/s	
	M_2	Re_2	M_2	Re_2	M_2	Re_2
0.3	0.18	19645	0.48	48527	0.93	85971
0.4	0.10	14646	0.25	36215	0.48	64559
0.5	0.06	11750	0.16	28989	0.29	51536
0.6	0.04	9780	0.11	24163	0.20	42967
0.7	0.03	8383	0.08	20705	0.14	36812
0.8	0.03	7335	0.06	18118	0.11	32216

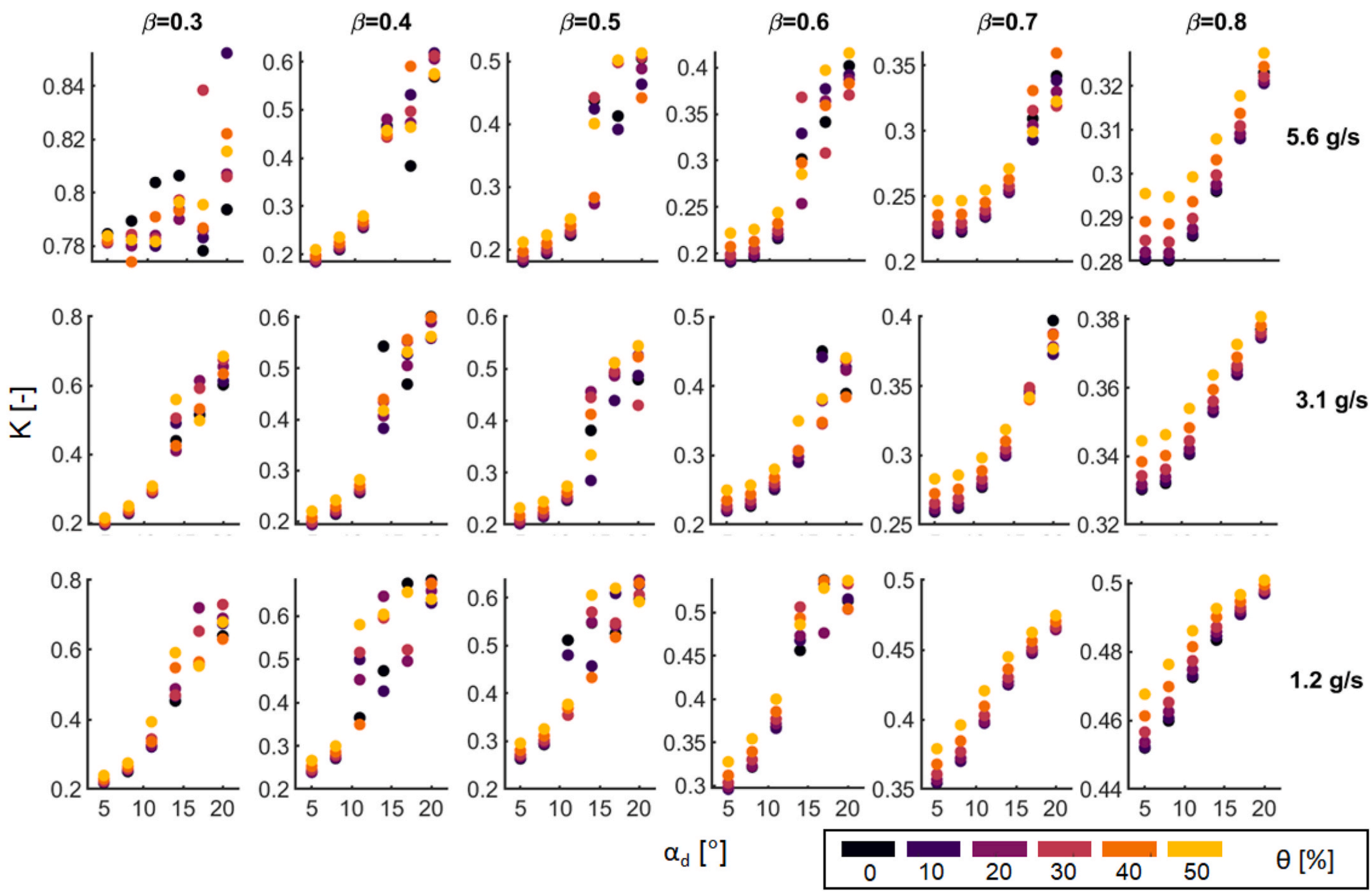

Fig. 7. Pressure Loss Coefficients obtained by varying mass flow, diameter ratio, truncation, and divergent angle.

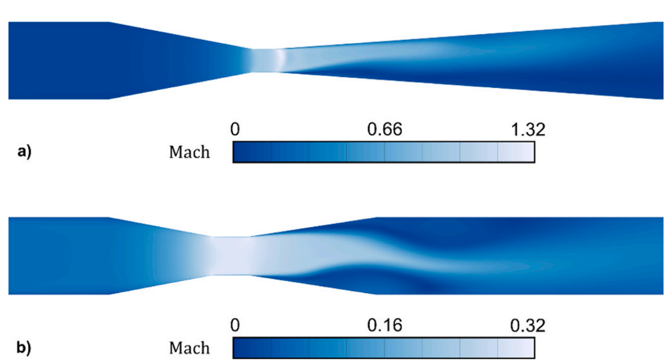

Fig. 8. Mach number contours of two high mass flow cases ($\dot{m} = 5.6$ g/ s). a) *Model 37*: $\beta = 0.3$, $\theta = 0$, $\alpha_d = 8°$. b) *Model 147*: $\beta = 0.5$, $\theta = 0$, $\alpha_d = 17°$.

same discussion regarding steadiness for Fig. 7 applies to the discharge coefficients shown in Fig. 9. In this case, as expected, the discharge coefficient can be considered independent of divergent angle and truncation. It is mainly influenced by the diameter ratio or throat Reynolds number [4], listed in Table 3. For instance, the differences caused by truncation and divergent angle are on the order of 0.1 % for the medium mass flow cases.

The sensitivity of mass flow calculation to pressure measured in *Plane 2* (P_2) is also calculated. This is presented in Fig. 10 as the relative error in mass flow for 1 Pa uncertainty in pressure measurement. The sensitivity to pressure P_2, increases with the diameter ratio, β. There is a negligible increase in mass flow error with increasing truncation ($<\sim1$ %).

3.3. Response surface modeling

A total of 90 design points were used as input for the regression to compute the response surface coefficients in Equation (8). The points were extracted from the medium mass flow case ($\dot{m} = 3.1$ g/s) with β ranging from 0.4 to 0.8, θ ranging from 0 to 0.5, and α_d ranging from 5° to 11°. The response surface, computed with an $R^2 = 0.992$, is found to be:

$$\widehat{K}(\boldsymbol{x}) = 3.45 \bullet 10^{-1} - 8.22 \bullet 10^{-1}\beta + 2.28 \bullet 10^{-2}\theta + 6.50 \bullet 10^{-3}\alpha_d - 5.72 \bullet 10^{-2}\beta\theta - 2.24 \bullet 10^{-2}\beta\alpha_d - 5.27 \bullet 10^{-4}\theta\alpha_d + 1.06\beta^2 + 1.25 \bullet 10^{-1}\theta^2 + 7.92 \bullet 10^{-4}\alpha_d^2 \tag{10}$$

This function allows for identifying the influence of these parameters on pressure loss. The results for eleven individuals at medium mass flow ($\dot{m} = 3.1$ g/s) are summarized in Table 4. The pressure loss coefficient tends to increase with diameter ratio (β), truncation (θ), and divergent angle (α_d). For the pressure loss coefficient (K), the pipe loss is not subtracted, hence, an increase in loss with diameter ratio is seen, which is consistent with the literature [7]. Fig. 11 shows the pressure loss ratio (ξ), for the cases without truncation and a divergent angle of 8°. The trend reveals that there is a sharp rise in pressure loss ratio for higher diameter ratios when the pipe loss is not subtracted. This is due to the pipe's major (friction) losses being higher than the Venturi expansion losses.

Further inspection of Equation (10) shows that the larger the diameter ratio, the less important the impact of truncation is on pressure loss. For instance, the difference in pressure loss coefficient between *Model 3* and *Model 33* is 15.2 %, and the difference between *Model 6* and *Model 36* is 4.3 %. Furthermore, with a larger diameter ratio, the effect of the

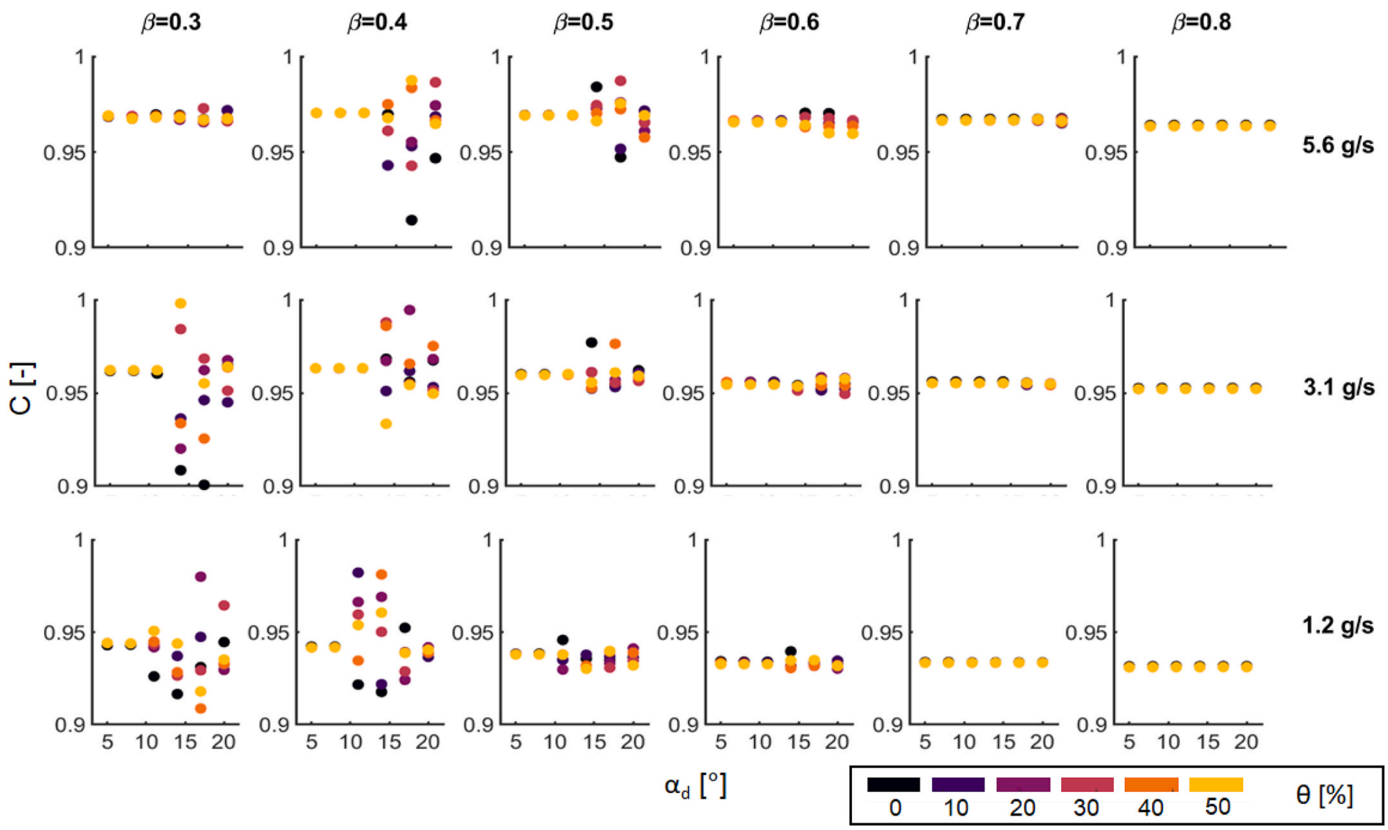

Fig. 9. Discharge Coefficient obtained by varying mass flow, diameter ratio, truncation, and divergent angle.

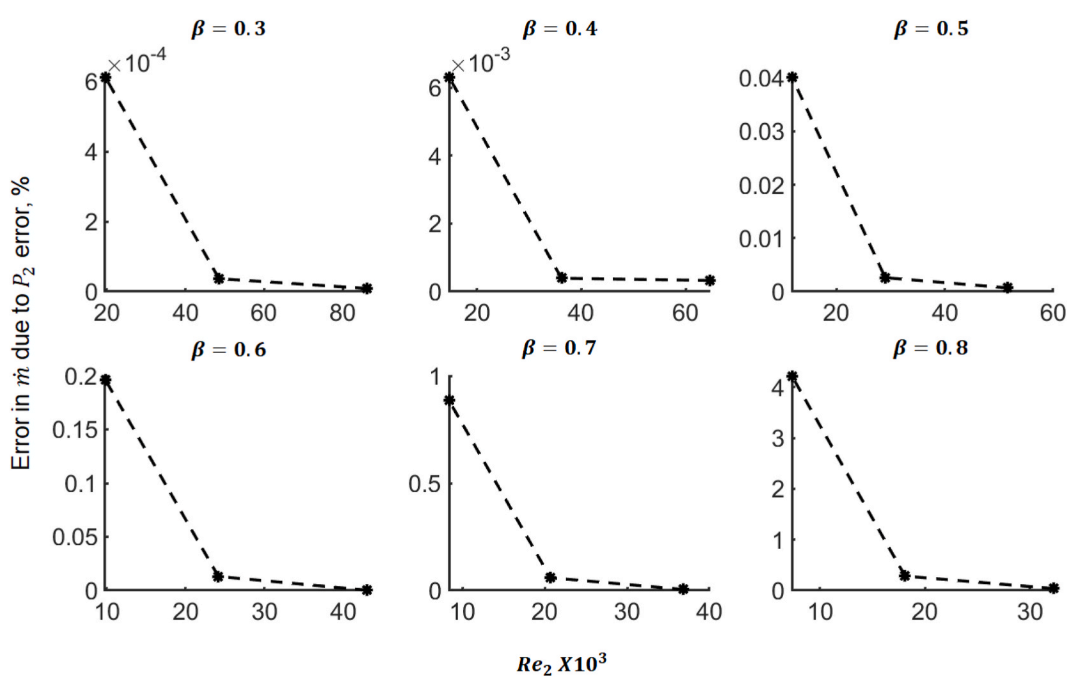

Fig. 10. Averaged relative error in mass flow calculation for 1 Pa uncertainty in pressure measurement of *Plane 2* pressure, P_2 vs Reynolds number at *Plane 2* for different diameter ratios.

divergent angle becomes less important, as observed in the $\beta = 0.8$ cases in Fig. 7, where there is little scatter present in the computed cases. This can also be observed by calculating the difference in pressure loss coefficient between *Model 3* and *Model 75,* which is 22.5 % while the difference between *Model 6* and *Model 78* is 3.1 %. The last parameter interaction is the divergent angle and the truncation, where a larger divergent angle results in a smaller effect of truncation. This is seen by comparing *Model 3* and *Model 33* with a 15.2 % difference in pressure loss coefficient and *Model 75* and *Model 105* with an 11 % difference in pressure loss coefficient.

Table 4
Medium mass flow ($\dot{m} = 3.1$ g/s) tabulated examples.

Model	β	θ	α_d	K	ξ	C
3	0.5	0	5°	0.202	0.164	0.960
33	0.5	0.5	5°	0.232	0.195	0.960
75	0.5	0	11°	0.247	0.210	0.960
105	0.5	0.5	11°	0.274	0.238	0.960
40	0.6	0	8°	0.227	0.156	0.956
58	0.6	0.3	8°	0.236	0.165	0.954
70	0.6	0.5	8°	0.257	0.189	0.954
6	0.8	0	5°	0.330	0.101	0.953
36	0.8	0.5	5°	0.344	0.124	0.952
78	0.8	0	11°	0.341	0.118	0.953
108	0.8	0.5	11°	0.354	0.140	0.952

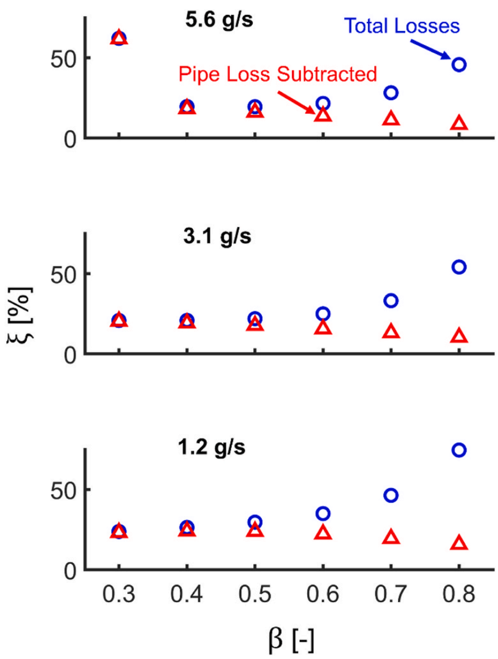

Fig. 11. Pressure Loss Ratio, including the pipe loss and with the pipe loss subtracted. For geometries with $\theta = 0$, $\alpha_d = 8°$.

A nonlinear increase in pressure loss with truncation (θ) is observed, which is also seen in Fig. 7. Hence, truncation should not be neglected during design. For example, the difference in pressure loss between *Model 40* and *Model 58 is* 3.9 %, while between *Model 40* and *Model 70 is* 13.4 %. Velocity and turbulent kinetic energy contours for the aforementioned 0 % and 50 % truncation cases are depicted in Fig. 12. In the truncated cases, the sudden expansion causes a recirculation region with a lower velocity magnitude, leading to a detached shear layer, thus increasing the turbulent kinetic energy being shed.

3.4. Optimal recovery cone angle

Though the pressure loss is monotonically increasing with the studied parameters, there still is an optimal truncation and divergent angle for a given recovery cone length. To find this optimum, the DOE results are transformed into a function of recovery cone length. Equation (7) is used to map $K(\beta,\theta,\alpha_d)$ to $K(\beta,\theta,L_d)$ for divergent angles from 5° to 11° and truncation ranging from 0 % to 50 %. Three quadratic response surfaces are fitted with Equation (8) for the cases $\beta = 0.4$, $\beta = 0.6$, and $\beta = 0.8$, yielding three functions $\widehat{K}_{\beta=0.4}(\theta,L_d)$, $\widehat{K}_{\beta=0.6}(\theta,L_d)$, and $\widehat{K}_{\beta=0.8}(\theta,L_d)$. The value of truncation (θ) which results in minimum pressure loss coefficient (K) for a given recovery cone length (L_d) is found. Table 5 shows the coefficient of determination (R^2) for the response surfaces at

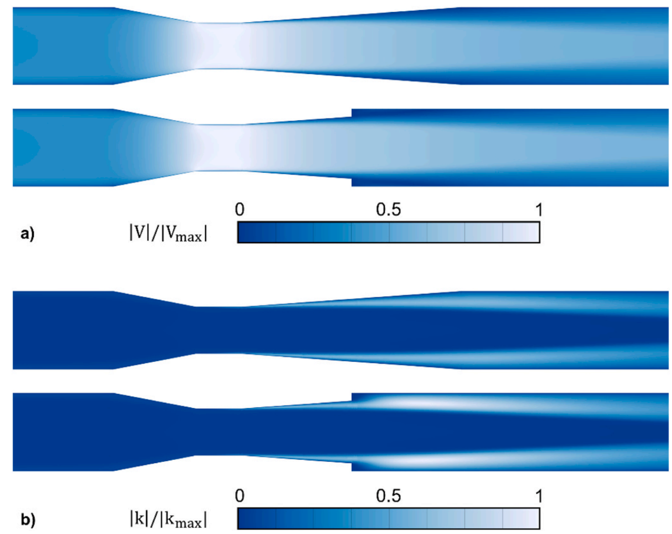

Fig. 12. Normalized velocity (a) and turbulent kinetic energy (b) contours of two medium mass flow cases ($\dot{m} = 3.1$ g/s). Geometries with $\beta = 0.6$, $\alpha_d = 8°$ and $\theta = 0$ (*Model 40*, top) or $\theta = 0.5$ (*Model 70*, bottom).

Table 5
Coefficient of determination (R^2) for the response surfaces generated.

	$\dot{m} = 1.231$ g/s	$\dot{m} = 3.130$ g/s	$\dot{m} = 5.628$ g/s
$\beta = 0.4$	0.7596	0.9797	0.9818
$\beta = 0.6$	0.9801	0.9770	0.9778
$\beta = 0.8$	0.9939	0.9812	0.9861

each corresponding mass flow condition. The minimum pressure loss coefficient is found for ten different lengths equispaced between the minimum and maximum lengths possible for the selected diameter ratios. Fig. 13 shows an example of the computations to find the minimum losses and highlights the tradeoff between truncation and divergent angle.

The optimum is a tradeoff between expansion losses due to truncation and losses due to separation at higher divergent angles. For small cone lengths, the required divergent angles are higher, with greater chances of separation. The optimizer prefers to gradually slow the flow down at a small angle and then suddenly expand to the full diameter. Since expansion losses are proportional to flow velocity, the total loss is reduced by slowing the flow before sudden expansion at the location of truncation rather than having the flow separate earlier with a larger divergent angle. When the cone length is longer and the risk of separation is smaller, the optimum avoids expansion losses due to truncation altogether. There is also an overall decrease in losses with the increase in recovery cone length.

The minimum values obtained for all the cases are depicted in Fig. 14. The trend observed is similar for all cases. Shorter recovery cones require larger divergent angles and high truncation to achieve the prescribed length. The truncation and divergent angle can be smaller for longer recovery cones to achieve minimum losses. Furthermore, for larger diameter ratios (i.e., $\beta = 0.8$), a smaller recovery angle is found to be optimal, and the truncation can be reduced due to the throat diameter being closer to the pipe diameter. This trend for the divergent angle is present for all three mass flow cases shown in Fig. 14. However, a larger truncation is allowed for the low mass flow case ($\dot{m} = 1.2$ g/s) since the effect of truncation is smaller at lower Reynolds numbers. This chart can be used as a reference to design compact Venturis.

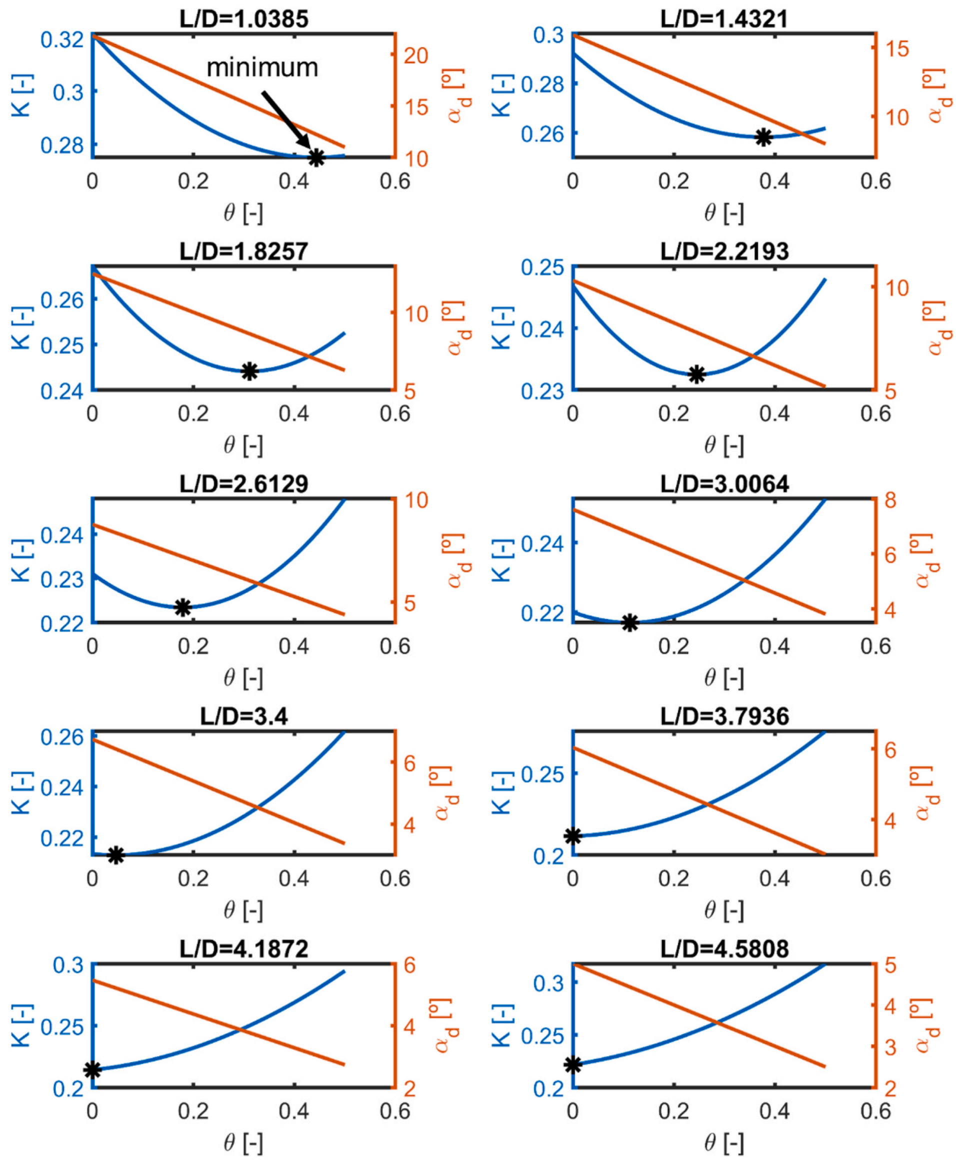

Fig. 13. Example of the procedure followed to find the minimum pressure loss coefficient for the medium mass flow case ($\dot{m} = 3.130$ g/s) and $\beta = 0.6$.

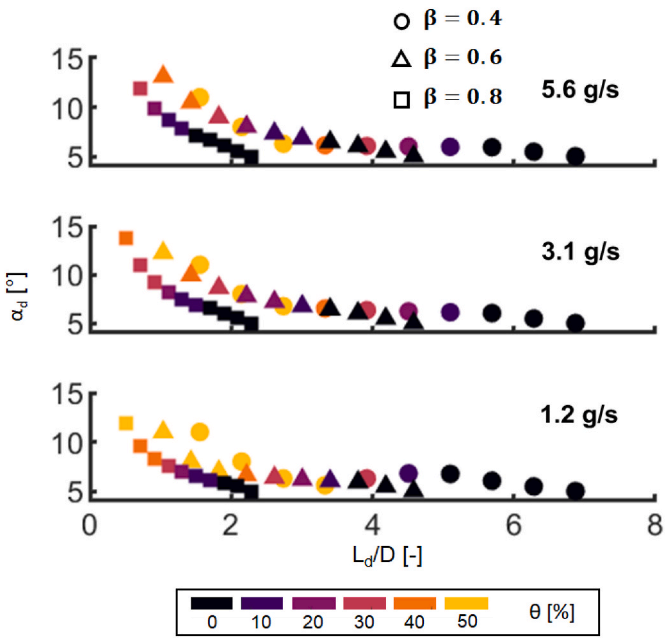

Fig. 14. Optimal divergent angle and truncation for a given recovery cone length.

4. Conclusions

Two hundred sixteen (216) Venturi flow meter geometries were carefully analyzed at three different mass flow conditions to assess the effect of the diameter ratio, truncation, and recovery cone angle. A truncated Venturi flow meter was manufactured and tested to validate the numerical setup at eleven different mass flow points. The influence of the different geometrical parameters on the pressure loss, discharge coefficient, and mass flow uncertainty were quantified. Design guidelines were provided for all experimentalists.

The study covered throat Reynolds numbers from $7 \bullet 10^3$ to $7 \bullet 10^4$. The results are applicable to turbulent flows in single-phase subsonic Venturi flow meters running full. The investigated geometrical parameters have an important effect on the pressure losses and strongly influence each other, so they should be considered together when designing a flow meter for a specific condition. Compact Venturis operating at low Reynolds exhibit high uncertainty levels. The error in mass flow measurement decreases with decreasing diameter ratio since the pressure differential increases. However, for small diameter ratios, divergent cone angles larger than 10° should be implemented carefully because flow peculiarities such as significant flow separation followed by unsteadiness may appear. In the cases where the space is constrained, it is recommended that the designers choose a less aggressive cone angle and truncate the Venturi according to the charts presented. There is an optimal value of truncation and divergent angle for a given Venturi length. The optimal angle and truncation both decrease with longer recovery cone lengths. This manuscript offers essential design guidelines for low Reynolds flow applications such as aerothermal testing and biomedicine.

Credit author statement

A. Rebassa: Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization. L. Bhatnagar: Conceptualization, Formal analysis, Investigation, Data Curation, Writing - Review & Editing, Visualization, Funding acquisition. G. Paniagua: Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was partially financed by the Shah Family Global Innovation Lab's 2020 Seed Grants Award toward "Affordable water meter at scale in India", Grant Number 2020_034_IND_06_08. The authors would like to acknowledge the Shah Family Global Innovation Lab and Wateraid India for their support. The authors would also like to acknowledge Ms. Miranda Leigh Daane's efforts in the experimental setup.

Nomenclature

English symbols

a	= Quadratic response surface coefficient
C	= Discharge coefficient
D	= Upstream diameter
d	= Throat diameter
K	= Pressure loss coefficient
k	= Turbulent Kinetic Energy
L	= Length
M	= Mach number
$\dot{m}$	= Mass flow
n	= Number of design variables
P	= Static pressure
R	= Specific gas constant or coefficient of determination
Re	= Reynolds number
S	= Sensitivity Coefficient
T	= Temperature
v	= Velocity
X	= Variable perturbed in the sensitivity coefficient
x	= design variable or coordinate in the x-direction

y^+ = Dimensionless wall distance

Greek symbols

α = Cone angle
β = Throat to upstream diameter ratio
γ = Specific heat ratio
Δ = Variation
ε = Expansibility Factor
θ = Truncation parameter
μ = Dynamic Viscosity
ν = Kinematic Viscosity
ξ = Pressure Loss Ratio
ρ = Density
τ = Throat to upstream pressure ratio

Subscripts

0 = Full-length Venturi
1 = Plane 1 (upstream)
2 = Plane 2 (throat)
CFD = Computational Fluid Dynamics
c = Convergent section
d = Divergent section
exp = Experimental model
in = Inlet plane
loss = Losses between inlet and outlet
o = Total quantity
out = Outlet plane
pipe = Pipe prior to venturi installation
t = Truncated
throat = Throat section

Appendix

Table A 1
Design of Experiment results for each flow meter model. The values listed are pressure loss coefficient with pipe loss (K), pressure loss ratio with pipe loss subtracted (ξ), and discharge coefficient (C).

Model	β	θ	α_d	$\dot{m}$ = 1.231 g/s			$\dot{m}$ = 3.130 g/s			$\dot{m}$ = 5.628 g/s		
				K	ξ	C	K	ξ	C	K	ξ	C
1	0.3	0	5	0.218	0.201	0.943	0.195	0.174	0.962	0.785	0.613	0.968
2	0.4	0	5	0.240	0.211	0.942	0.193	0.172	0.963	0.185	0.160	0.970
3	0.5	0	5	0.263	0.209	0.938	0.202	0.164	0.960	0.181	0.147	0.969
4	0.6	0	5	0.297	0.196	0.934	0.220	0.148	0.956	0.191	0.131	0.966
5	0.7	0	5	0.354	0.176	0.934	0.259	0.128	0.956	0.222	0.111	0.967
6	0.8	0	5	0.452	0.146	0.932	0.330	0.101	0.953	0.280	0.085	0.964
7	0.3	0.1	5	0.220	0.203	0.944	0.197	0.176	0.962	0.781	0.610	0.969
8	0.4	0.1	5	0.239	0.210	0.942	0.194	0.172	0.963	0.184	0.160	0.970
9	0.5	0.1	5	0.265	0.211	0.938	0.202	0.164	0.960	0.182	0.149	0.969
10	0.6	0.1	5	0.296	0.196	0.933	0.220	0.148	0.956	0.192	0.132	0.966
11	0.7	0.1	5	0.355	0.176	0.933	0.260	0.129	0.955	0.223	0.112	0.966
12	0.8	0.1	5	0.452	0.147	0.931	0.331	0.102	0.952	0.281	0.086	0.963
13	0.3	0.2	5	0.221	0.204	0.944	0.198	0.176	0.962	0.782	0.610	0.969
14	0.4	0.2	5	0.241	0.212	0.942	0.196	0.174	0.963	0.186	0.161	0.970
15	0.5	0.2	5	0.268	0.214	0.938	0.205	0.167	0.960	0.184	0.151	0.969
16	0.6	0.2	5	0.298	0.199	0.933	0.222	0.151	0.956	0.194	0.134	0.966
17	0.7	0.2	5	0.357	0.178	0.933	0.262	0.131	0.955	0.225	0.114	0.966
18	0.8	0.2	5	0.454	0.149	0.931	0.332	0.103	0.952	0.282	0.088	0.963
19	0.3	0.3	5	0.224	0.206	0.944	0.200	0.179	0.962	0.781	0.610	0.969
20	0.4	0.3	5	0.245	0.216	0.942	0.200	0.178	0.963	0.190	0.165	0.970
21	0.5	0.3	5	0.272	0.218	0.938	0.209	0.171	0.960	0.189	0.156	0.969
22	0.6	0.3	5	0.303	0.204	0.933	0.227	0.156	0.956	0.199	0.139	0.966
23	0.7	0.3	5	0.361	0.184	0.933	0.266	0.136	0.955	0.229	0.119	0.966
24	0.8	0.3	5	0.457	0.154	0.931	0.334	0.108	0.952	0.285	0.092	0.963
25	0.3	0.4	5	0.229	0.212	0.944	0.206	0.183	0.962	0.782	0.611	0.969
26	0.4	0.4	5	0.253	0.224	0.942	0.207	0.185	0.963	0.197	0.172	0.970
27	0.5	0.4	5	0.281	0.227	0.938	0.218	0.180	0.960	0.197	0.164	0.969
28	0.6	0.4	5	0.312	0.213	0.933	0.235	0.165	0.956	0.208	0.149	0.966
29	0.7	0.4	5	0.368	0.193	0.933	0.272	0.145	0.955	0.236	0.128	0.966
30	0.8	0.4	5	0.461	0.161	0.931	0.338	0.114	0.952	0.289	0.099	0.963
31	0.3	0.5	5	0.239	0.221	0.944	0.215	0.192	0.962	0.784	0.613	0.969
32	0.4	0.5	5	0.266	0.237	0.942	0.220	0.198	0.963	0.210	0.183	0.970

(continued on next page)

Table A 1 (*continued*)

Model	β	θ	α_d	$\dot{m}$ = 1.231 g/s			$\dot{m}$ = 3.130 g/s			$\dot{m}$ = 5.628 g/s		
				K	ξ	C	K	ξ	C	K	ξ	C
33	0.5	0.5	5	0.296	0.242	0.938	0.232	0.195	0.960	0.212	0.179	0.969
34	0.6	0.5	5	0.327	0.230	0.933	0.250	0.181	0.954	0.222	0.165	0.966
35	0.7	0.5	5	0.379	0.206	0.933	0.283	0.158	0.955	0.247	0.142	0.966
36	0.8	0.5	5	0.468	0.172	0.931	0.344	0.124	0.952	0.296	0.110	0.963
37	0.3	0	8	0.250	0.232	0.943	0.228	0.204	0.962	0.789	0.618	0.967
38	0.4	0	8	0.271	0.241	0.942	0.214	0.192	0.963	0.210	0.184	0.970
39	0.5	0	8	0.293	0.240	0.938	0.215	0.178	0.960	0.194	0.161	0.969
40	0.6	0	8	0.321	0.223	0.934	0.227	0.156	0.956	0.197	0.137	0.966
41	0.7	0	8	0.370	0.195	0.934	0.262	0.131	0.956	0.223	0.112	0.967
42	0.8	0	8	0.460	0.159	0.932	0.332	0.104	0.953	0.280	0.085	0.964
43	0.3	0.1	8	0.254	0.236	0.944	0.229	0.205	0.962	0.783	0.613	0.969
44	0.4	0.1	8	0.272	0.243	0.942	0.216	0.194	0.963	0.212	0.185	0.970
45	0.5	0.1	8	0.295	0.241	0.938	0.216	0.178	0.960	0.195	0.162	0.969
46	0.6	0.1	8	0.322	0.224	0.933	0.229	0.158	0.954	0.198	0.138	0.966
47	0.7	0.1	8	0.370	0.195	0.933	0.263	0.132	0.955	0.224	0.113	0.967
48	0.8	0.1	8	0.461	0.160	0.931	0.333	0.105	0.952	0.281	0.086	0.964
49	0.3	0.2	8	0.256	0.237	0.944	0.231	0.207	0.962	0.780	0.609	0.968
50	0.4	0.2	8	0.274	0.245	0.942	0.218	0.196	0.963	0.213	0.187	0.970
51	0.5	0.2	8	0.297	0.243	0.938	0.218	0.180	0.960	0.197	0.164	0.969
52	0.6	0.2	8	0.322	0.225	0.933	0.230	0.159	0.956	0.200	0.140	0.966
53	0.7	0.2	8	0.372	0.198	0.933	0.265	0.135	0.955	0.226	0.116	0.967
54	0.8	0.2	8	0.463	0.163	0.931	0.334	0.107	0.952	0.282	0.088	0.964
55	0.3	0.3	8	0.259	0.240	0.944	0.235	0.210	0.962	0.784	0.613	0.969
56	0.4	0.3	8	0.279	0.249	0.942	0.222	0.200	0.963	0.217	0.190	0.970
57	0.5	0.3	8	0.302	0.249	0.938	0.223	0.185	0.960	0.202	0.169	0.969
58	0.6	0.3	8	0.330	0.233	0.933	0.236	0.165	0.954	0.205	0.146	0.966
59	0.7	0.3	8	0.377	0.204	0.933	0.269	0.140	0.955	0.230	0.121	0.967
60	0.8	0.3	8	0.465	0.168	0.931	0.336	0.111	0.952	0.285	0.092	0.964
61	0.3	0.4	8	0.265	0.246	0.944	0.240	0.215	0.962	0.774	0.602	0.968
62	0.4	0.4	8	0.286	0.256	0.942	0.229	0.207	0.963	0.224	0.197	0.970
63	0.5	0.4	8	0.311	0.258	0.938	0.231	0.194	0.960	0.210	0.177	0.969
64	0.6	0.4	8	0.339	0.243	0.933	0.244	0.175	0.954	0.213	0.155	0.966
65	0.7	0.4	8	0.385	0.213	0.933	0.276	0.149	0.955	0.236	0.129	0.967
66	0.8	0.4	8	0.470	0.175	0.931	0.340	0.117	0.952	0.289	0.098	0.964
67	0.3	0.5	8	0.275	0.255	0.944	0.249	0.224	0.962	0.782	0.610	0.967
68	0.4	0.5	8	0.300	0.269	0.942	0.242	0.219	0.963	0.236	0.208	0.970
69	0.5	0.5	8	0.326	0.272	0.938	0.244	0.207	0.960	0.223	0.191	0.969
70	0.6	0.5	8	0.354	0.259	0.933	0.257	0.189	0.954	0.226	0.169	0.966
71	0.7	0.5	8	0.396	0.228	0.933	0.286	0.161	0.955	0.247	0.142	0.966
72	0.8	0.5	8	0.476	0.186	0.931	0.346	0.127	0.952	0.295	0.108	0.964
73	0.3	0	11	0.322	0.300	0.926	0.293	0.263	0.960	0.804	0.634	0.970
74	0.4	0	11	0.365	0.336	0.922	0.256	0.233	0.963	0.256	0.226	0.970
75	0.5	0	11	0.511	0.460	0.946	0.247	0.210	0.960	0.223	0.190	0.969
76	0.6	0	11	0.368	0.274	0.934	0.251	0.182	0.956	0.216	0.159	0.966
77	0.7	0	11	0.398	0.231	0.934	0.277	0.150	0.956	0.234	0.127	0.967
78	0.8	0	11	0.473	0.179	0.932	0.341	0.118	0.953	0.286	0.094	0.964
79	0.3	0.1	11	0.321	0.300	0.942	0.288	0.259	0.962	0.780	0.609	0.968
80	0.4	0.1	11	0.500	0.457	0.982	0.258	0.235	0.963	0.257	0.227	0.970
81	0.5	0.1	11	0.480	0.429	0.935	0.249	0.212	0.960	0.225	0.192	0.969
82	0.6	0.1	11	0.366	0.272	0.933	0.252	0.183	0.956	0.218	0.160	0.966
83	0.7	0.1	11	0.397	0.229	0.933	0.278	0.151	0.955	0.235	0.127	0.967
84	0.8	0.1	11	0.473	0.180	0.931	0.341	0.118	0.952	0.286	0.095	0.964
85	0.3	0.2	11	0.326	0.304	0.942	0.289	0.261	0.962	0.784	0.613	0.968
86	0.4	0.2	11	0.453	0.416	0.966	0.260	0.237	0.963	0.259	0.229	0.970
87	0.5	0.2	11	0.369	0.317	0.930	0.251	0.214	0.960	0.227	0.194	0.969
88	0.6	0.2	11	0.372	0.278	0.933	0.256	0.188	0.954	0.221	0.163	0.966
89	0.7	0.2	11	0.398	0.231	0.933	0.279	0.153	0.955	0.236	0.129	0.967
90	0.8	0.2	11	0.475	0.183	0.931	0.342	0.120	0.952	0.288	0.097	0.964
91	0.3	0.3	11	0.344	0.321	0.944	0.292	0.263	0.962	0.783	0.611	0.969
92	0.4	0.3	11	0.516	0.482	0.960	0.264	0.240	0.963	0.262	0.232	0.970
93	0.5	0.3	11	0.355	0.302	0.938	0.254	0.217	0.960	0.229	0.197	0.969
94	0.6	0.3	11	0.377	0.284	0.933	0.260	0.193	0.954	0.225	0.168	0.966
95	0.7	0.3	11	0.403	0.236	0.933	0.283	0.158	0.955	0.240	0.134	0.967
96	0.8	0.3	11	0.477	0.187	0.931	0.345	0.124	0.952	0.290	0.100	0.964
97	0.3	0.4	11	0.336	0.314	0.945	0.298	0.269	0.962	0.791	0.621	0.969
98	0.4	0.4	11	0.349	0.317	0.935	0.270	0.247	0.963	0.269	0.239	0.970
99	0.5	0.4	11	0.369	0.316	0.938	0.263	0.226	0.960	0.239	0.206	0.969
100	0.6	0.4	11	0.385	0.293	0.933	0.268	0.201	0.954	0.233	0.176	0.966
101	0.7	0.4	11	0.410	0.245	0.933	0.289	0.165	0.955	0.245	0.141	0.967
102	0.8	0.4	11	0.481	0.194	0.931	0.348	0.130	0.952	0.294	0.107	0.964
103	0.3	0.5	11	0.393	0.368	0.951	0.307	0.277	0.962	0.782	0.611	0.968
104	0.4	0.5	11	0.580	0.547	0.954	0.282	0.258	0.963	0.280	0.249	0.970
105	0.5	0.5	11	0.377	0.325	0.938	0.274	0.238	0.960	0.249	0.217	0.969
106	0.6	0.5	11	0.400	0.309	0.933	0.280	0.214	0.954	0.244	0.189	0.966

(*continued on next page*)

Table A 1 (*continued*)

Model	β	θ	α_d	$\dot{m} = 1.231$ g/s			$\dot{m} = 3.130$ g/s			$\dot{m} = 5.628$ g/s		
				K	ξ	C	K	ξ	C	K	ξ	C
107	0.7	0.5	11	0.421	0.259	0.933	0.298	0.177	0.955	0.255	0.152	0.967
108	0.8	0.5	11	0.486	0.202	0.931	0.354	0.140	0.952	0.299	0.116	0.964
109	0.3	0	14	0.453	0.424	0.917	0.439	0.391	0.909	0.806	0.636	0.969
110	0.4	0	14	0.474	0.441	0.918	0.543	0.514	0.968	0.466	0.427	0.970
111	0.5	0	14	0.547	0.496	0.936	0.381	0.346	0.977	0.438	0.406	0.984
112	0.6	0	14	0.456	0.369	0.940	0.290	0.226	0.954	0.302	0.251	0.970
113	0.7	0	14	0.427	0.267	0.934	0.300	0.179	0.956	0.254	0.151	0.967
114	0.8	0	14	0.483	0.197	0.932	0.353	0.138	0.953	0.296	0.110	0.964
115	0.3	0.1	14	0.467	0.426	0.937	0.491	0.447	0.936	0.794	0.623	0.969
116	0.4	0.1	14	0.426	0.394	0.922	0.382	0.357	0.951	0.462	0.418	0.943
117	0.5	0.1	14	0.457	0.405	0.938	0.285	0.250	0.952	0.424	0.392	0.974
118	0.6	0.1	14	0.468	0.383	0.932	0.290	0.226	0.953	0.329	0.282	0.963
119	0.7	0.1	14	0.425	0.264	0.933	0.300	0.179	0.955	0.253	0.150	0.967
120	0.8	0.1	14	0.484	0.199	0.931	0.353	0.138	0.952	0.296	0.111	0.964
121	0.3	0.2	14	0.488	0.464	0.898	0.410	0.367	0.920	0.790	0.618	0.967
122	0.4	0.2	14	0.646	0.615	0.969	0.407	0.378	0.967	0.480	0.442	1.002
123	0.5	0.2	14	0.549	0.499	0.931	0.456	0.423	0.961	0.273	0.241	0.973
124	0.6	0.2	14	0.473	0.389	0.932	0.298	0.235	0.953	0.254	0.200	0.964
125	0.7	0.2	14	0.427	0.266	0.933	0.301	0.181	0.955	0.255	0.152	0.967
126	0.8	0.2	14	0.486	0.201	0.931	0.354	0.140	0.952	0.298	0.113	0.964
127	0.3	0.3	14	0.470	0.442	0.926	0.505	0.463	0.984	0.797	0.629	0.968
128	0.4	0.3	14	0.596	0.558	0.950	0.435	0.408	0.988	0.444	0.404	0.961
129	0.5	0.3	14	0.570	0.521	0.931	0.445	0.412	0.961	0.443	0.411	0.974
130	0.6	0.3	14	0.506	0.424	0.931	0.307	0.245	0.951	0.368	0.324	0.968
131	0.7	0.3	14	0.430	0.271	0.933	0.305	0.186	0.955	0.258	0.156	0.967
132	0.8	0.3	14	0.487	0.203	0.931	0.356	0.143	0.952	0.300	0.117	0.964
133	0.3	0.4	14	0.548	0.522	0.928	0.425	0.382	0.934	0.793	0.623	0.969
134	0.4	0.4	14	0.601	0.562	0.981	0.440	0.411	0.986	0.449	0.409	0.975
135	0.5	0.4	14	0.433	0.382	0.932	0.412	0.379	0.953	0.283	0.251	0.970
136	0.6	0.4	14	0.493	0.411	0.931	0.306	0.244	0.954	0.298	0.248	0.963
137	0.7	0.4	14	0.436	0.278	0.933	0.310	0.192	0.955	0.263	0.163	0.967
138	0.8	0.4	14	0.490	0.208	0.931	0.359	0.148	0.952	0.303	0.122	0.964
139	0.3	0.5	14	0.592	0.566	0.944	0.559	0.516	0.998	0.797	0.626	0.968
140	0.4	0.5	14	0.604	0.565	0.960	0.417	0.390	0.933	0.457	0.416	0.968
141	0.5	0.5	14	0.605	0.557	0.930	0.334	0.300	0.956	0.401	0.370	0.966
142	0.6	0.5	14	0.486	0.401	0.935	0.350	0.291	0.954	0.285	0.235	0.964
143	0.7	0.5	14	0.445	0.289	0.933	0.319	0.203	0.955	0.271	0.173	0.967
144	0.8	0.5	14	0.493	0.212	0.931	0.364	0.155	0.952	0.308	0.130	0.964
145	0.3	0	17	0.557	0.507	0.931	0.514	0.465	0.901	0.778	0.605	0.967
146	0.4	0	17	0.676	0.636	0.952	0.469	0.438	0.956	0.384	0.342	0.914
147	0.5	0	17	0.525	0.475	0.935	0.493	0.461	0.954	0.413	0.381	0.947
148	0.6	0	17	0.537	0.458	0.934	0.450	0.402	0.952	0.341	0.295	0.970
149	0.7	0	17	0.450	0.295	0.934	0.346	0.238	0.955	0.309	0.221	0.967
150	0.8	0	17	0.491	0.209	0.932	0.366	0.158	0.953	0.309	0.132	0.964
151	0.3	0.1	17	0.721	0.687	0.947	0.529	0.468	0.946	0.783	0.610	0.967
152	0.4	0.1	17	0.656	0.616	0.939	0.528	0.499	0.962	0.531	0.483	0.953
153	0.5	0.1	17	0.609	0.560	0.937	0.439	0.406	0.953	0.392	0.361	0.952
154	0.6	0.1	17	0.533	0.454	0.932	0.442	0.392	0.951	0.377	0.334	0.965
155	0.7	0.1	17	0.447	0.292	0.933	0.344	0.236	0.954	0.293	0.201	0.967
156	0.8	0.1	17	0.491	0.209	0.931	0.364	0.156	0.952	0.308	0.130	0.964
157	0.3	0.2	17	0.720	0.686	0.980	0.614	0.568	0.962	0.786	0.613	0.966
158	0.4	0.2	17	0.496	0.456	0.924	0.506	0.481	0.994	0.472	0.430	0.955
159	0.5	0.2	17	0.542	0.492	0.933	0.486	0.455	0.957	0.498	0.468	0.976
160	0.6	0.2	17	0.476	0.392	0.933	0.379	0.322	0.958	0.364	0.320	0.965
161	0.7	0.2	17	0.449	0.294	0.933	0.345	0.236	0.955	0.304	0.215	0.966
162	0.8	0.2	17	0.492	0.211	0.931	0.365	0.157	0.952	0.309	0.132	0.964
163	0.3	0.3	17	0.653	0.617	0.929	0.592	0.539	0.968	0.838	0.672	0.973
164	0.4	0.3	17	0.522	0.487	0.929	0.553	0.521	0.955	0.497	0.449	0.943
165	0.5	0.3	17	0.547	0.497	0.931	0.495	0.463	0.955	0.499	0.469	0.987
166	0.6	0.3	17	0.536	0.457	0.932	0.346	0.286	0.954	0.308	0.259	0.968
167	0.7	0.3	17	0.451	0.297	0.933	0.349	0.241	0.955	0.315	0.229	0.967
168	0.8	0.3	17	0.493	0.213	0.931	0.366	0.160	0.952	0.311	0.135	0.964
169	0.3	0.4	17	0.565	0.534	0.909	0.532	0.487	0.926	0.787	0.614	0.966
170	0.4	0.4	17	0.656	0.616	0.939	0.557	0.523	0.966	0.590	0.544	0.984
171	0.5	0.4	17	0.517	0.466	0.939	0.512	0.480	0.976	0.502	0.471	0.972
172	0.6	0.4	17	0.536	0.457	0.932	0.348	0.289	0.954	0.359	0.315	0.963
173	0.7	0.4	17	0.456	0.303	0.933	0.340	0.230	0.955	0.331	0.248	0.967
174	0.8	0.4	17	0.495	0.215	0.931	0.369	0.164	0.952	0.314	0.139	0.964
175	0.3	0.5	17	0.554	0.512	0.918	0.498	0.452	0.955	0.796	0.624	0.967
176	0.4	0.5	17	0.655	0.616	0.939	0.532	0.502	0.954	0.464	0.423	0.988
177	0.5	0.5	17	0.620	0.571	0.940	0.511	0.480	0.961	0.502	0.472	0.975
178	0.6	0.5	17	0.528	0.447	0.935	0.382	0.325	0.957	0.397	0.356	0.960
179	0.7	0.5	17	0.462	0.311	0.933	0.342	0.232	0.955	0.299	0.208	0.967
180	0.8	0.5	17	0.497	0.219	0.931	0.373	0.170	0.952	0.318	0.146	0.964

(*continued on next page*)

Table A 1 (*continued*)

Model	β	θ	α_d	$\dot{m}$ = 1.231 g/s			$\dot{m}$ = 3.130 g/s			$\dot{m}$ = 5.628 g/s		
				K	ξ	C	K	ξ	C	K	ξ	C
181	0.3	0	20	0.640	0.601	0.945	0.602	0.549	0.965	0.794	0.622	0.968
182	0.4	0	20	0.683	0.642	0.939	0.601	0.569	0.967	0.568	0.518	0.947
183	0.5	0	20	0.595	0.545	0.936	0.479	0.447	0.962	0.505	0.474	0.965
184	0.6	0	20	0.513	0.433	0.933	0.389	0.335	0.955	0.402	0.362	0.966
185	0.7	0	20	0.466	0.316	0.934	0.397	0.302	0.955	0.342	0.262	0.968
186	0.8	0	20	0.497	0.219	0.932	0.377	0.177	0.953	0.323	0.154	0.964
187	0.3	0.1	20	0.675	0.639	0.933	0.613	0.548	0.945	0.852	0.684	0.972
188	0.4	0.1	20	0.630	0.592	0.937	0.559	0.527	0.953	0.618	0.571	0.969
189	0.5	0.1	20	0.627	0.578	0.936	0.487	0.455	0.961	0.464	0.433	0.972
190	0.6	0.1	20	0.515	0.434	0.935	0.427	0.376	0.953	0.392	0.351	0.966
191	0.7	0.1	20	0.465	0.314	0.933	0.373	0.272	0.955	0.338	0.258	0.965
192	0.8	0.1	20	0.497	0.219	0.931	0.375	0.173	0.952	0.321	0.150	0.964
193	0.3	0.2	20	0.690	0.633	0.930	0.655	0.595	0.968	0.807	0.636	0.967
194	0.4	0.2	20	0.658	0.618	0.942	0.591	0.557	0.968	0.606	0.555	0.974
195	0.5	0.2	20	0.637	0.588	0.941	0.525	0.494	0.958	0.488	0.458	0.961
196	0.6	0.2	20	0.504	0.422	0.930	0.423	0.371	0.958	0.387	0.345	0.966
197	0.7	0.2	20	0.464	0.314	0.933	0.378	0.278	0.955	0.330	0.247	0.967
198	0.8	0.2	20	0.497	0.220	0.931	0.375	0.174	0.952	0.321	0.151	0.964
199	0.3	0.3	20	0.730	0.693	0.964	0.676	0.618	0.951	0.806	0.634	0.966
200	0.4	0.3	20	0.676	0.636	0.941	0.561	0.528	0.951	0.612	0.563	0.986
201	0.5	0.3	20	0.606	0.557	0.935	0.430	0.396	0.956	0.507	0.477	0.965
202	0.6	0.3	20	0.533	0.454	0.932	0.438	0.388	0.949	0.371	0.327	0.966
203	0.7	0.3	20	0.466	0.316	0.933	0.388	0.290	0.954	0.319	0.233	0.967
204	0.8	0.3	20	0.498	0.221	0.931	0.376	0.176	0.952	0.322	0.153	0.964
205	0.3	0.4	20	0.631	0.594	0.933	0.634	0.581	0.964	0.822	0.652	0.968
206	0.4	0.4	20	0.674	0.634	0.939	0.600	0.566	0.975	0.572	0.522	0.967
207	0.5	0.4	20	0.630	0.581	0.939	0.523	0.492	0.959	0.442	0.411	0.958
208	0.6	0.4	20	0.504	0.422	0.932	0.384	0.329	0.954	0.383	0.341	0.964
209	0.7	0.4	20	0.470	0.321	0.933	0.387	0.289	0.954	0.359	0.284	0.966
210	0.8	0.4	20	0.499	0.223	0.931	0.378	0.179	0.952	0.324	0.157	0.964
211	0.3	0.5	20	0.679	0.641	0.935	0.685	0.628	0.964	0.816	0.644	0.967
212	0.4	0.5	20	0.639	0.600	0.940	0.563	0.531	0.950	0.575	0.525	0.965
213	0.5	0.5	20	0.591	0.543	0.932	0.544	0.515	0.959	0.513	0.483	0.969
214	0.6	0.5	20	0.537	0.458	0.932	0.441	0.390	0.958	0.416	0.377	0.960
215	0.7	0.5	20	0.475	0.326	0.933	0.377	0.276	0.955	0.322	0.238	0.966
216	0.8	0.5	20	0.501	0.225	0.931	0.381	0.183	0.952	0.327	0.162	0.964

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